QQQ Weekly Credit Spread Strategy: Spec v1

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Purpose and fit

The strategy sells defined-risk weekly credit spreads on QQQ to earn in sideways and low-volatility markets, where the NDX regime trader mostly sits in cash.

- **Complement, not duplicate** - The NDX trader follows trends with TQQQ/SQQQ and earns in sustained moves. Option selling earns when the market stays inside a range, so the two should pay in different conditions.
- **Separate account** - It runs in its own brokerage account with a fixed base of $40-50K. Realized profits above the base are swept into the NDX trader every 6-12 months.
- **Origin** - The idea came from two observations. The NDX trader sits in cash through choppy markets, and index options are usually priced for bigger moves than actually happen. Selling defined-risk spreads on the side the trend is moving away from aims to collect that gap while the trader waits.

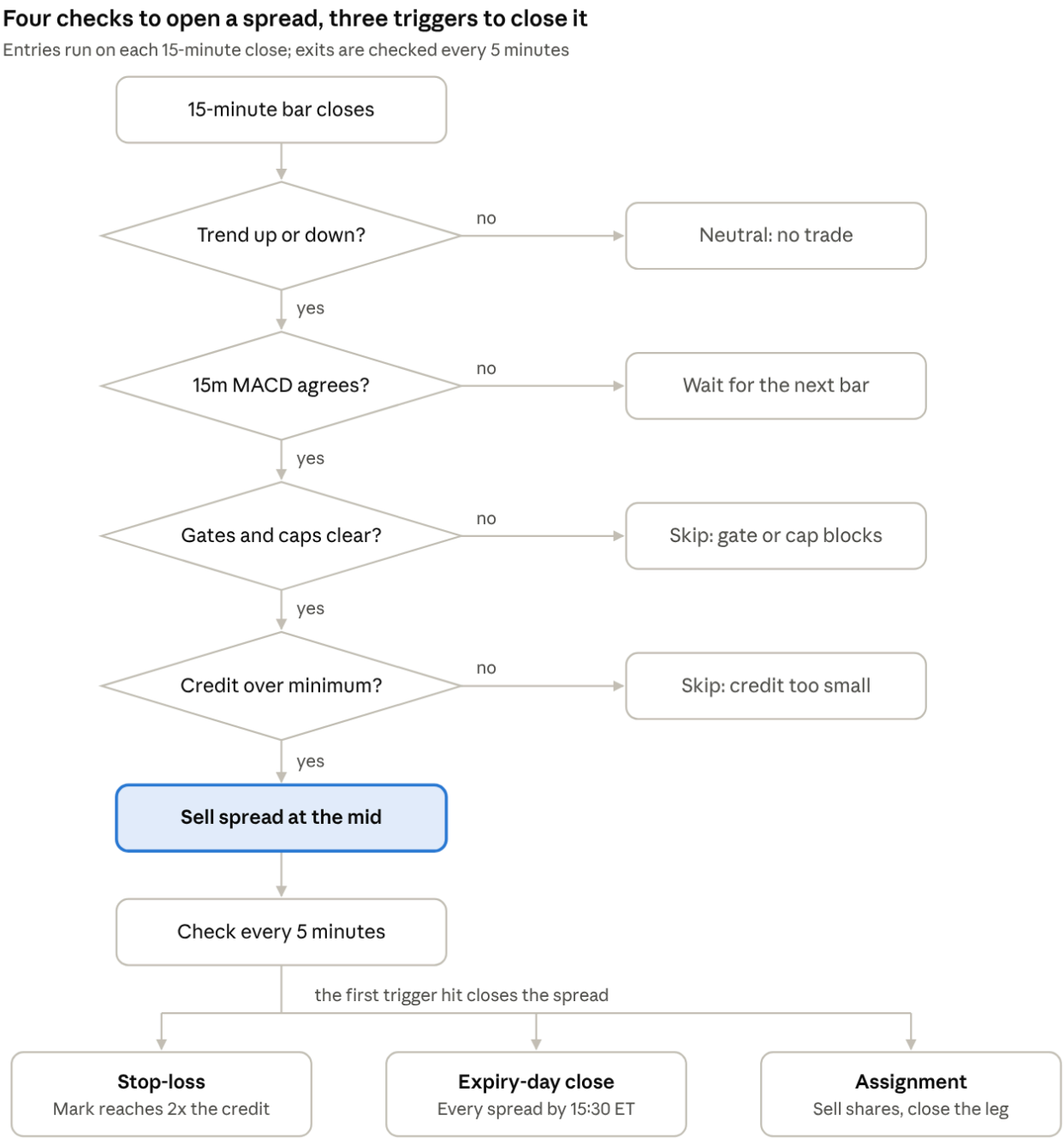
Design principles

Four principles drive every rule in this spec.

1. **Defined risk on every position** - Each short option is paired with a long option further out of the money. No naked short options, at any size.
2. **Stops handle normal moves; the long leg handles gaps** - A stop fills at the next available price, so an overnight gap can jump straight past it. The long leg caps the loss however far the market gaps.
3. **Rules first, machine learning later** - The core is simple rules that are easy to test and trust. Machine learning is added only as an optional filter on top, once there is enough trade history to train it.
4. **Judged on the combined portfolio** - The strategy is accepted only if it improves the NDX trader and options book together, not on standalone returns alone.

How a trade flows

A spread opens only after four checks pass on a 15-minute close, and it closes on the first of three exit triggers.



Life of a trade: 4 entry checks, 3 exit triggers

A failed check changes nothing until the next bar. Gates block new entries but never close open spreads.

Instrument and expiry

The strategy trades QQQ options at about 7 days to expiry, the most liquid options on the Nasdaq-100.

Item	Choice	Why
Underlying	QQQ	Tightest quotes on the Nasdaq-100, the same index the NDX trader runs on
Expiry	Nearest to 7 calendar days (window 5-9)	Collects most of the weekly time decay while leaving time to manage the trade; QQQ lists an expiry every weekday
Contract	100 shares, American-style, settled in shares	Early assignment and expiry-day pin risk must be managed (see Exit rules)
Alternative	XND (Nasdaq-100 Micro Index options)	Cash-settled and European-style, so no assignment; checked in Phase 2 for quote width

One assigned QQQ contract delivers 100 shares, a position larger than the whole account. The engine therefore never holds a short leg into the close on expiry day.

Entry rules

An entry fires when the 15-minute trend state points up or down and the 15-minute MACD histogram agrees. The spread is sold on the side the trend is moving away from.

Rule	Definition
Trend lines	Time-series moving average (linear-regression endpoint) of the close over 14 and 26 bars (TSF14, TSF26), on QQQ 15-minute bars
Trend state	Up: TSF14 above TSF26 and close above TSF14. Down: TSF14 below TSF26 and close below TSF14. Anything else is neutral: no trade
MACD gate	MACD(12, 26, 9) histogram on 15-minute bars must agree: above zero for up, below zero for down. If not, wait
Side	Up state sells a put spread; down state sells a call spread
Reinforcement	The 1-hour trend state agrees with the 15-minute state. Logged, not a gate
High-confidence tier	15m and 1h states agree, 1h MACD agrees, and 1h RSI(14) is extreme in the trend's direction (above 80 when up, below 20 when down). Logged in v1, tested as a size-up in Phase 2
Bars	Closed, clock-aligned bars only, with at least 30 days of 5-minute history for indicator warm-up
Entry window	Each 15-minute close from 9:45 to 15:45 ET
Caps	At most 5 open spreads per side; at least 30 minutes between new entries on the same side

The trigger is the trend state, not a fresh crossover. Waiting for a fresh crossover would miss most of a trend.

Not in v1: longer-term bias filters and support/resistance levels. The NDX trader's regime is tested as the bias filter in Phase 2.

Strike selection and structure

The short leg is picked by delta, the long leg sits a fixed width further out, and a trade with too little credit is skipped.

Element	v1 default	Phase 2 tests
Short leg	Strike nearest 10 delta on the chosen side	5, 10, 15, 20 delta
Long leg	Same expiry, $5 further out of the money	$3, $5, $10
Minimum credit	Skip if net credit is below $0.30 per share ($30 per spread)	$0.15 to $0.60
Order	One combo order for both legs, limit at the mid, never legged in	Fill rules: mid, mid plus 1 cent steps
Size	1 spread per signal	Size-up for the high-confidence tier

- **Delta sets the distance** - Picking the short strike by delta adjusts automatically: it moves further out when volatility is high or expiry is further away.
- **Minimum credit** - Skipping trades that pay too little keeps costs from eating the premium and avoids near-worthless options, which carry the worst gap risk for the least reward. The level is set in Phase 2, where net P&L turns positive.
- **Support/resistance check** - Never sell a strike inside a confirmed support or resistance level. A 1-hour Supertrend version is tested in Phase 2.

Exit rules

Each spread closes on the first trigger hit, checked every 5 minutes, and nothing is held into the close on expiry day.

Trigger	Rule	Why
Stop-loss	Close when the spread's mark reaches 2x the credit received. Mark = the higher of the spread's mid price and its intrinsic value	Caps a losing trade at about the size of the credit, while leaving room for normal noise
Expiry-day close	Close every spread expiring today by 15:30 ET, at any price	Removes pin risk and after-hours assignment
Profit-take	None in v1. Phase 2 tests closing at 50% and 75% of the credit captured	Holding collects more time decay; closing early cuts risk
Early assignment	Check positions at the open and after every order. If a leg was assigned, sell the delivered shares and close the other leg at once, then alert	One contract delivers 100 shares, more than the account
Ex-dividend	Close any in-the-money short call the day before a QQQ ex-dividend date	The usual cause of early call assignment

Exits go out as one combo order at a marketable limit, up to the natural price. The 5-minute check matches the NDX trader's stop loop.

Risk limits and gates

Losses are capped per spread, across open spreads and on account drawdown; gates only block new entries and never force exits.

Limit	v1 setting
Max loss per spread	About 1% of the account ($5 wide on QQQ), never above 2%

Limit	v1 setting
Open spreads	5 per side, 10 in total
Total open max loss	Sum of max losses on open spreads at most 20% of the account
Drawdown overlay	At -10% from peak equity, halve the per-side cap. At -15%, halt new entries until reset by hand. Peak and halt persist across days, as in the NDX trader

Gate	Rule	Status
Volatility	No new entries when 20-day realized volatility is above 35%, the NDX trader's own filter	v1
Term structure	No new entries when VIX is above VIX3M, a sign of short-term market stress	v1
Weekend	No new entries after 13:00 ET on Fridays, to avoid opening positions into weekend gaps	Phase 2 test
Events	No new entries on the day before or the day of FOMC decisions and CPI releases	Phase 2 test

Live operations

The options engine reuses the NDX trader's proven building blocks and adds only the option-specific parts.

Reused from the NDX trader

- Automatic reconnects, a watchdog that restarts a stuck process, heartbeat files and an external dead-man's switch
- Email reports and error alerts, persisted risk state, and scheduling on the same server

New for options

- Option chain lookup (expiries and strikes), then live quotes and Greeks for strikes near the target delta
- Spread orders for both legs at once: limit at the mid, stepped toward the natural price, cancelled and retried if unfilled
- Reconciliation every check: broker positions mapped back to spreads, with alerts for assigned or orphaned legs
- Every order and position query names its account, so the two strategies can never cross
- A daily email after the close: open spreads, credits, open max loss, P&L, gate status and any assignment

Phase 2 research plan

The strategy goes to build only if it passes all four tests on data it was not tuned on.

1. **Timing adds value** - It beats a mechanical baseline: the same spreads, stop and sizing, entered weekly with no signal (one put spread version, one iron condor version). If it does not, the simpler baseline ships instead.
2. **It earns while the trader is flat** - Net P&L is positive on days the NDX trader holds cash, sliced by the trader's daily state.
3. **The combined portfolio improves** - The NDX trader plus the options book has a better return-to-drawdown ratio than the trader alone. The worst combined days, such as a crash out of an uptrend,

are listed and survivable.

4. **It survives real costs** - Net P&L stays positive with real bid/ask and broker commissions, and still positive when every fill pays the full bid/ask spread.

Method - Tune on the earlier history, then judge the final versions once on a sealed holdout of the most recent 12 months. Use the longest history the data allows, covering at least the 2020 crash, the 2022 bear market and the April 2025 tariff shock. Stops are checked every 5 minutes, as live.

Area	Variants tested
Strikes	Short leg 5, 10, 15, 20 delta; width $3, $5, $10; minimum credit $0.15 to $0.60
Expiry	5, 7 and 10 days
Exits	Profit-take none, 50%, 75%; stop on spread mark vs short-leg mark; exit when the trend state flips
Gates	Volatility threshold; VIX vs VIX3M; Friday cutoff; FOMC and CPI blackout
Signal	High-confidence size-up; NDX regime as a side tilt; support/resistance strike filter
Overlay	ML entry veto, trained walk-forward on the backtest's own trades

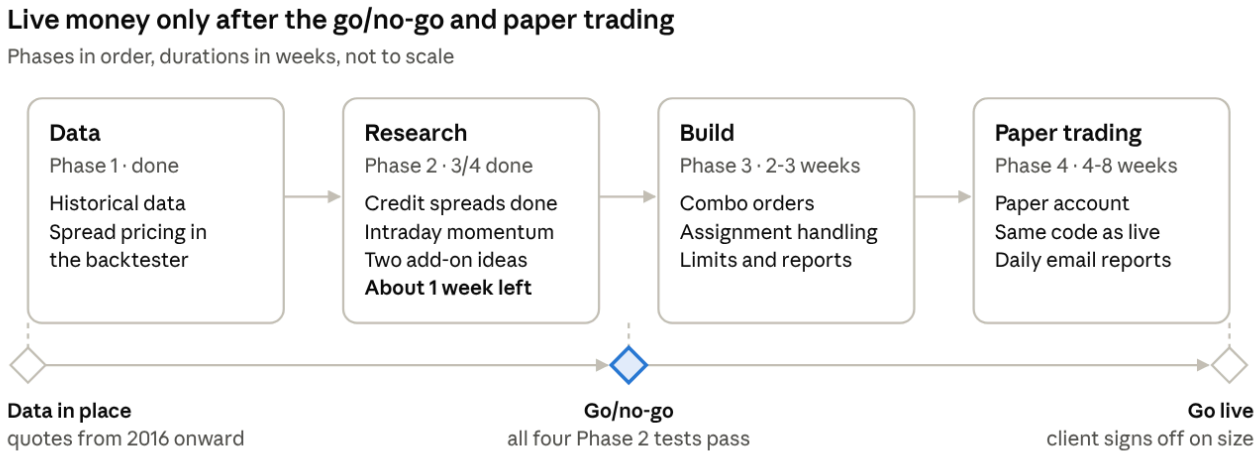
Metrics - Net P&L, annual return on the account, max drawdown, worst day and week, win rate, average win vs loss, P&L by side and by trader state, and the combined portfolio's return-to-drawdown ratio.

Data

Backtests use minute-by-minute historical quotes for every QQQ option strike from 2016 onward, with bid, ask, delta and implied volatility, plus daily VIX, VIX3M and NDX closes. This data is already in place.

Phases and timeline

The data phase is done, and research is about three-quarters done with roughly one more week to go. That week finishes testing intraday momentum and the two add-on ideas from the backtest report: a short-volatility ETF and an opening-gap fade. Build and paper trading then take roughly 6-11 weeks, and real money moves only after the Phase 2 go/no-go and paper trading.



Delivery roadmap: 4 phases, 3 gates

Going live waits for the client to choose the size.

Appendix: input variables

Variable	Live source	Backtest source	Used in
QQQ 15-minute and 1-hour bars	Broker market data	Historical minute data	Trend state, MACD, RSI
TSF14, TSF26	Computed	Computed	Entry trigger
MACD(12, 26, 9) histogram, 15m and 1h	Computed	Computed	Entry gate, high-confidence tier
RSI(14), 1h	Computed	Computed	High-confidence tier
Option bid/ask and delta	Broker option quotes	Historical option quotes	Strike selection, stop mark, fills
VIX, VIX3M	Market data	Daily closes	Term-structure gate
20-day realized volatility	NDX daily closes	NDX daily closes	Volatility gate
NDX regime and trader state	NDX trader logs	NDX trader backtest	Phase 2 tilt and P&L slicing
Account equity and peak	Broker account	Backtest ledger	Sizing, drawdown overlay
FOMC and CPI dates	Published calendars	Published calendars	Event blackout (Phase 2)
QQQ ex-dividend dates	Published fund data	Historical dividend list	Early-assignment rule